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Bayesian Probability Quiz

Review Learning Objectives

Instructions

coach

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Assignment details

Submitted	Attempts
Nov 18, 2:01 AM IST	Unlimited

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1. What does Bayes' rule primarily allow us to do? **1 / 1 point**

- (●) Update the probability of a hypothesis based on new evidence.
- ( ) Calculate the probability of observing data given a hypothesis.
- ( ) Estimate the initial probability of an event.
- ( ) Determine the likelihood of a hypothesis given the data.

Correct
Correct! Bayes' rule is fundamentally about updating the probability of a hypothesis as new evidence is acquired.

2. Which term in Bayes' rule represents the initial belief in the hypothesis before any data is observed? **1 / 1 point**

- ( ) Marginal probability
- (●) Prior probability
- ( ) Posterior probability
- ( ) Likelihood

Correct
Correct! The prior probability represents the initial degree of belief in the hypothesis before any data is observed.

3. Suppose you wish to determine if a coin is fair (meaning that heads and tails are equally likely outcomes). You believe that the chance the coin is fair is more likely (suppose 0.66) than not (0.34). You perform an experiment and flip the coin 5 times, all of which turn up heads. Using Bayes' rule, what would you estimate to be the probability that the coin is fair given the data you have collected? **1 / 1 point**

- (●) 0.27
- ( ) There is not enough information to calculate this.
- ( ) 0.5
- ( ) 0.03125

Correct
Correct! The probability of the data given that the coin is fair is 1/32. The probability of the data under both scenarios is approximately 0.0769.